Objects of Thought

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When do our thoughts begin to have the generative structure characteristic of fully developed human thought? This talk considers infants' representations of OBJECTS as a strong test case for an early language of thought.

Empirical evidence

Infants' representations of objects are characterized by:

i. **Featural content:** Infants can and do represent the surface features of objects, but they do not always do so. Features can be "forgotten" discretely.
   Citations: Applin & Kibbe, 2020; Cheng & Kibbe, 2024; Káldy & Leslie, 2003, 2005; Kibbe & Leslie, 2011; 2013; 2016; Kibbe & Applin, 2022
ii. **Abstract content:** Infants represent an object's kind without representing its surface features.
   Citations: Kibbe & Leslie, 2019; Kibbe & Stahl, 2023a; Pomiechowska & Gliga, 2022
iii. **Nothing but mere existence:** Infants representations of objects can contain no descriptive content at all about its surface features, kind, or physical properties (i.e., contra Spelke, object representations need not contain physical info)
   Citations: Kibbe & Leslie, 2011; Kibbe & Applin, 2022; Kibbe & Feigenson, 2016
iv. **Not dependent on perceptual access to objects:** Infants represent objects whose existence they must infer; representations of objects are not dependent on sensory input
   Citations: Cheng & Kibbe, under review; Kibbe & Feigenson, 2015
v. **Supports logical operators:** Infants' object representations support "exclusive or" for discrete, mutually exclusive content.
   Citations: Cheng & Kibbe, 2024; Turan-Küçük & Kibbe, 2024, 2025, 2026a, 2026b; Turan-Küçük, et al., 2026; see also Brody et al., 2026; see also Mazalik & Halberda; Cesana-Arlotti and colleagues

Theoretical claim

- Sparse, abstract initial structure for object representations – the bare *index* – which points to an object in a location
- Can be triggered by input from object files and/or property maps, but can be elicited even if objects are not directly observed
- Only the object's existence in a location is represented necessarily; *nothing else* need be represented
- Information about the object's identity or other properties (features, kinds, "stands-for" relations, etc., etc.) can optionally be linked to the index, but it doesn't have to be
- Index representations function as bare demonstratives in infants' (and indeed adults') language of thought – "THAT" – which means indexes can be inputs into early-emerging propositions like CAUSE, BELIEVES, etc.

Conclusion: We find evidence for generative, structured, abstract thought in infancy in the way infants represent and think about objects. This early structure provides the seed from more sophisticated knowledge can develop.